

Negative Exponent LawQuestion Bank - Additional 3.04 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 6, rewrite each expression as an equivalent expression with positive exponents. Do not evaluate.**

1. $\frac{2^{-8}}{2^{-3}}$

$$2^{-5} = \frac{1}{2^5} \quad (\text{or} \quad \frac{2^3}{2^8})$$

2. $-7 \cdot 2^{-4}$

$$-\frac{7}{2^4}$$

3. $\frac{1}{10^{-3}}$

$$10^3$$

4. $(-4)^{-5}$

$$\frac{1}{(-4)^5}$$

5. 9^{-6}

$$\frac{1}{9^6}$$

6. $\frac{1}{(-3)^{-7}}$

$$(-3)^7$$

For questions 7 and 8, apply the negative exponent law, then the product or quotient of powers law, and evaluate.

7. Evaluate each expression.

a. $(-7)^{-6} \cdot (-7)^4$

$$(-7)^{-2} = \frac{1}{49}$$

b. $8^{-5} \cdot 8^5$

$$8^0 = 1$$

c. $5^6 \cdot 5^{-3}$

$$5^3 = 125$$

8. Evaluate each expression.

a. $(-6)^{-8} \div (-6)^{-5}$

$$(-6)^{-3} = -\frac{1}{216}$$

b. $3^{-5} \div 3^2$

$$3^{-7} = \frac{1}{2,187}$$

c. $5^0 \div 5^{-2}$

$$5^2 = 25$$

Opposite or reciprocal.

9. Four expressions are shown: -10 , 10^{-1} , $(-10)^{-1}$, -10^{-1} .

a. Give the value of each one.

$$-10; 1/10; -1/10; -1/10$$

b. Which of the four is the reciprocal of 10, and which is the opposite of 10?

$$10^{-1} \text{ is the reciprocal; } -10 \text{ is the opposite.}$$

Working backwards.

10. Find the value of n that makes each statement true.

a. $3^n = \frac{1}{81}$

$$n = -4$$

b. $10^n = 0.00001$

$$n = -5$$

c. $\frac{6^2}{6^n} = 6^5$

$$n = -3$$

$$2 - n = 5.$$

11. Which is the value of 3^{-4} ?

☒ A. $\frac{1}{81}$

☐ B. $\frac{1}{12}$

☐ C. -81

☐ D. -12

A

12. Select ALL of the expressions whose value is $\frac{1}{81}$.

☐ A. 3^{-3}

☒ B. 3^{-4}

☐ C. -3^4

☒ D. 9^{-2}

☒ E. $\frac{3^{-6}}{3^{-2}}$

☒ F. $\left(\frac{1}{3}\right)^4$

B, D, E and F

E: $-6 - (-2) = -4$.